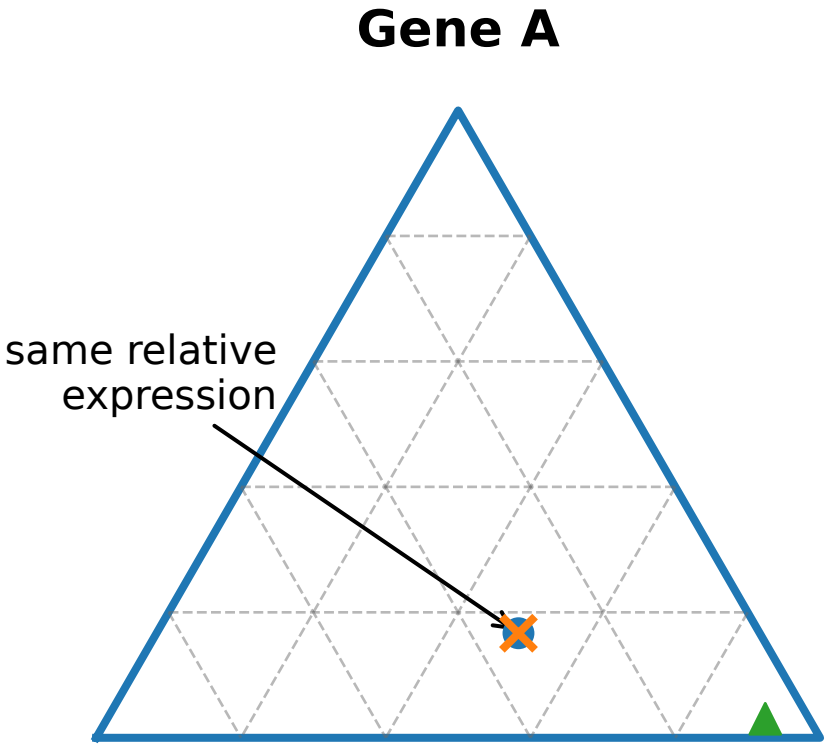
1. Gene expression in CLR space



Cell 1: (10, 20, 30)

Cell 2: (100, 200, 300)

Cell 3: (10, 20, 300)

$$z_{ig} = \log \frac{x_{ig}}{g(\mathbf{x}_i)}$$

$$g(\mathbf{x}_i) = \left(\prod_g x_{ig} \right)^{1/G}$$